# Evidence-Supported ASO Catalog

An attributed orbital catalog for Space Data Network

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## 1 Summary

Space Data Network (SDN) builds a catalog of anthropogenic space objects (ASOs) from public and login-only sources. It keeps every provider's product as an attributed record and, object by object, publishes the solution with the best measured accuracy for the intended use, with its uncertainty and lineage. Accuracy is measured against independent precise orbits, under plans fixed before the test data were read[1].

| Object class | Catalog product | Measured (median) |
|---|---|---|
| GPS satellites | Analysis-centre orbit, propagated | 3.5 cm at issue, 30 m a day later |
| Satellites with laser ranging | High-precision orbit propagation (HPOP) fitted to the ranges | 7 m at the fit, 42 m after 7 days |
| Satellites with operator ephemerides | The operator's ephemeris | Public element sets (OMMs) lie 0.6–5 km from it |
| Objects known only from public OMMs | SGP4 on the OMM | 0.3–1.6 km at issue |

For most objects the only public input is the OMM, and the catalog takes it as it is. What the catalog adds is a measured account of each input's error, and screening that acts on it safely:

1. **Measured error as covariance.** An orbit propagated from an OMM's epoch state carries a covariance equal to the error measured for its orbit regime and element-set age. That error is measured against independent orbits in every regime that has them: GNSS analysis-centre orbits, laser ranging, LEO precise orbits, operator ephemerides and radio-occultation receivers. A covariance is labeled calibrated only where held-out data confirm it (section 6).
2. **SDN's own atmosphere.** Drag uses JB2008 or NRLMSISE-00 with an open correction in the form of the Space Force's Dynamic Calibration of the Atmosphere, solved from public operator states. It explains 25–40 % of the density variation measured by satellites that did not calibrate it; 1–3-day position error is unchanged (ratios 0.98–1.02) (section 5.4).
3. **Maneuvers.** A detector reads each object's element-set history for steps in orbital energy and plane, and reports each maneuver's time and size. On held-out data it found 22 of 25 ISS burns and all 6 GPS maneuvers in precise orbits; geostationary station-keeping is harder, 4 of 11 (section 5.5).
4. **Better start states where data allow.** HPOP's force models agree with Orekit within 1.6 cm over a day. Fitted to laser ranges, HPOP stays 20–60 times better than SGP4 to 7 days; fitted to an arc of corrected OMMs, it halves SGP4's error to two days in GPS (section 5.3).
5. **Screening that stays safe when the covariance is large.** A large covariance spreads a collision's probability thin: on real OMM errors, a probability threshold of 1e-4 missed 2.7–16 % of true collisions. Where held-out data confirm the covariance, screening on the uncertainty region missed 0–0.4 %, at the cost of more false alerts, and every event also carries the covariance-free maximum probability[50]. Large uncertainty makes the screen cautious, not blind.

Three results go beyond published work:

1. GPS element-set correction fails on an unseen orbital plane (section 5.3).
2. A density correction of the kind the Space Force's drag model uses, solved from Planet's public states, is scored on satellites that did not calibrate it (section 5.4).
3. Consecutive OMM errors are strongly correlated, and four published covariance constructions are overconfident (section 6).

Pre-registered hypotheses that failed are reported as not supported.

## 2 Evidence model and records

An orbit solution is an inference with five contributors:

1. The object's physics
2. The dynamics model
3. The observations
4. The measurement model
5. The inference method

A provider's ephemeris or element set already embeds the provider's models and inference, so the catalog keeps it as an attributed provider claim, not as an observation. Agreement between products counts as independent confirmation only when their lineage is independent; unknown lineage is recorded as unknown.

Orbit data sit at four processing levels, by analogy with Earth-science data levels. A higher level is more processed, not better: it inherits every flaw below it, and re-epoching adds no evidence. Every source record declares its level.

| Level | Content | SDN inputs |
|---|---|---|
| 0 | Raw sensor data | Detector counts, signal samples, images |
| 1 | Calibrated, time-tagged measurements | Laser normal points; radio Doppler; optical angles |
| 2 | Estimated states and ephemerides | Operator ephemerides; GNSS and laser-ranging precise orbits; Vimpel elements; ILRS predictions |
| 3 | Theory-specific mean elements fitted to level 1 or 2 | OMMs from Space-Track and SpaceMapper; CelesTrak's supplemental GP; SDN's SGP4 companions |

A source record keeps the provider, native identifier, edition, SHA-256 of the bytes, product type, epoch, time scale and frame; observation time, solution epoch, issue time and retrieval time stay distinct. A solution record specifies:

| Field | Content |
|---|---|
| State and validity | State or ephemeris, epoch, applicable interval |
| Evidence and lineage | Source records, their levels and dependencies, editions |
| Dynamics | Force model, environment inputs, parameters estimated, considered or assumed |
| Measurement model | Sensor geometry, timing, calibration, biases; unknown when a provider does not state it |
| Inference | Estimator or selection rule, weights, priors, software version |
| Uncertainty | Covariance or none, its source and its calibration label |

Hashes establish byte identity and signatures establish attribution; physical validity rests on the measurements of sections 5 to 7. Identical inputs, module binaries and policy reproduce the same decisions.

## 3 Sources

The inventory[2] lists 97 public and login-only sources in eight groups, with access, terms, latency and use. Results draw on every source an experiment read; files are copied only where their terms allow, and the credits close this paper.

| Group | Sources | Role |
|---|---|---|
| A. Precise orbits | IGS and ESA GNSS orbits; ILRS orbits and normal points; Sentinel-1, Swarm, GRACE-FO and CryoSat-2 precise orbits | Truth; the GNSS and laser-ranged products |
| B. Operator ephemerides | SpaceX Starlink (with covariance), Planet, NASA ISS, CMSE Tiangong, Intelsat; ILRS predictions | Operator products; drag and maneuver evidence |
| C. Catalogs and observations | Space-Track GP history (OMM) and SATCAT; JSC Vimpel; SpaceMapper AOE; SatNOGS Doppler | Element sets; other providers' catalogs; public observations |
| D. Density | Swarm and GRACE-FO densities; CHAMP and GRACE-A densities with the High Accuracy Satellite Drag Model (HASDM)[3] | Density truth; the HASDM comparison |
| E. Space weather | Space Environment Technologies' (SET) JB2008 indices; GFZ Kp, ap and F10.7; NOAA SWPC forecasts | Model drivers, as issued |
| F. Earth orientation, gravity, ephemerides | IERS EOP; EGM2008, EGM96; FES2004; DE440 | Frames and dynamics |
| G. Physical properties | GCAT; IGS satellite metadata | Priors for drag and radiation parameters |
| H. The High Accuracy Catalog (HAC) and HASDM | Special Perturbations (SP) vectors; real-time HASDM | The comparison target (section 11) |

SpaceMapper's AOE catalog (channel KY) holds 19,197 element sets that the provider attributes to its own sensor network[4]. AOE rewrites them every 3 hours; SDN takes a snapshot daily from the public web pages. SpaceMapper states no license, so the sets are cited, not copied. Doppler read from SatNOGS waterfall images made the ISS orbit 2.7–10.7 times worse than SGP4 E6.

## 4 Normalization and model dispatch

*Vimpel osculating elements*

Vimpel's HEO product gives osculating elements in J2000[5]. The standard perifocal rotation turns them into a J2000 state at the same epoch[6], with $\mu = 398{,}600.4418$ km$^3$/s$^2$, under the identifier `vimpel:<id>` and no invented NORAD number[7]. Velocity differenced from 600 s position samples departs from the analytic velocity by up to 1.1 km/s, so it serves only as a diagnostic.

*The handoff contract*

An OMM holds SGP4 mean elements, not osculating ones. The catalog evaluates SGP4 at zero elapsed time, transforms position and velocity from TEME to GCRF at the same instant, and starts HPOP from that state[8]. SGP4's drag term B* becomes the ballistic parameter

$$B = \frac{C_D A}{m} = \frac{2B^*}{\rho_0}$$

with $\rho_0$ from Spacetrack Report No. 3[9]; 0.01 m²/kg stands in where $B^* \leq 0$, and $Cr \cdot A/m$ is 0.02 m²/kg[10]. The handoff is valid; whether it helps is measured in section 5.3.

*Levels of measurement of OMM elements*

OMM fields mix nominal (catalog number), ordinal (element-set number), interval (epoch), ratio (mean motion) and circular (node, perigee, mean anomaly) scales, so the pipeline computes on SGP4-evaluated states, never on raw fields. The legacy two-line format quantizes angles to 0.0001°, about 12 m along track near 6,900 km: a resolution, not an accuracy.

*Dispatch, frames and time*

Each record is routed by its declared theory: mean elements to SGP4, osculating elements to the two-body conversion, states and ephemerides to frame transforms. Osculating elements offered to SGP4 are refused as a model mismatch. Frames follow the IERS Conventions (2010): IAU 2006/2000A, CIO-based, through ERFA, with IERS Earth orientation[11]. TEME, EME2000, GCRF and ITRF stay distinct; epochs are UTC and integration runs on TT.

## 5 Products by object class

Each row is one method: its originators, our measured result (median 3D position error unless stated) and the experiment. Every experiment froze its plan before reading its test window[1].

*5.1 GNSS and laser-ranged satellites*

| Method | Prior art | Measured | Evidence |
|---|---|---|---|
| Newest ultra-rapid analysis-centre orbit at issue, propagated numerically beyond | IGS products[12] | GPS, 32 satellites: 3.5 cm at issue, 336 m at 7 days; the latest public OMM 1.5–1.7 km | E3 |
| HPOP fitted to ILRS laser normal points | Laser-ranging orbit determination[13] | Starlette, Stella, LARETS, WESTPAC, LARES: 7 m at the fit and 42 m at 7 days, against SGP4's 472 m and 685 m. One pass beats the OMM for a day; four passes beat it at every horizon to 7 days | E6 |

*5.2 Operator- and provider-ephemeris objects*

| Method | Prior art | Measured | Evidence |
|---|---|---|---|
| Operator ephemeris, within its span | Operator orbit determination | GNSS analysis centres' predictions (IGS for GPS, ESA for GLONASS): 0.1 m from final orbits at 0–12 h | E4 |
| SGP4 on the latest public OMM, against operator ephemerides | Dyreby et al. 2025[14]; Jankovic 2026[15] | At the file's start: Planet 0.59 km, ISS 0.82 km, GPS 1.23 km, Starlink 2.87 km, Intelsat 4.79 km; Starlink 13.8 km at 24 h | E4 |
| Vimpel's state refined to Vimpel's ephemeris by one six-state correction | Weighted least squares[16] | Withheld RMS 6.04–10.71 km before, 52.8–69.9 m after, on three 4-hour arcs: consistency with one provider, not accuracy | R77 |

*5.3 Objects with public OMMs only*

| Method | Prior art | Measured | Evidence |
|---|---|---|---|
| SGP4 on the latest public OMM | Hoots and Roehrich 1980[9]; Vallado et al. 2006[8] | At issue: 0.31 km (laser-ranged LEO) to 1.57 km (GPS); by horizon in section 11 | E3 |
| Along-track correction at epoch (GPS) | Ly et al. 2020: 65 %[17]; Hallgarten La Casta and Amato 2025[18] | RMS at epoch 2.04 → 0.79 km (61 % [53, 67]); the single-model form 8 %; worse than SGP4 by 7 days (2.55 against 1.88 km); its covariance fails the coverage gate. On an orbital plane outside the fit: 6 % [2, 10] | E1 |
| HPOP from the OMM's epoch state | Vallado et al. 2006[8]; Jankovic 2026[15] | Equal to SGP4 at epoch, then larger as the OMM's start-state error grows: at 24 h, 1.7 (GPS) to 11 (laser-ranged LEO) times SGP4's error; larger in all four regimes (GPS, LEO with precise orbits, laser-ranged LEO and MEO) by 72 h | E7 |
| HPOP fitted to an arc of OMM epoch states | Levit and Marshall 2011[19] | Better than SGP4 only at short horizons: 0.32–0.51 of its error to 12 h (laser-ranged LEO), 0.36–0.53 to 48 h on E1-corrected GPS sets; undecided beyond | E7 |

The start state sets the limit. Over 6 days, HPOP leaves 0.2 km along track from a GPS satellite's IGS state and 8.4 km from its OMM epoch state (medians of 508 arcs)[10]. Laser ranges fix the start state, and the prediction then stays 20 to 60 times better than SGP4 to 7 days E6.

*5.4 Drag regime*

Density is scored by $\sigma$, the spread of ln(observed/model density) on satellites outside the calibration; prediction by the 3-day position error relative to JB2008's (below 1 is better).

| Method | Prior art | Measured | Evidence |
|---|---|---|---|
| JB2008 with SET's indices | Bowman et al. 2008[20] | $\sigma$ 0.155 (Swarm, GRACE-FO, 2026); 0.148 on CHAMP and GRACE-A in 2003 and 2008, where HASDM scores 0.078 | E5 |
| Temperature correction from accelerometer and precise-orbit densities, a stand-in for HASDM's calibration | Storz et al. 2005[21]; Doornbos 2012[22] | $\sigma$ 0.056, 64 % [57, 70] lower; 0.112 on HASDM's windows; forecast gain 9 % for one day; 3-day ratio 1.04 [0.96, 1.10]; inputs arrive 49–92 days late (E8) | E5 |
| Temperature correction from the OMM decay of 48 catalogued objects | Picone et al. 2005[23]; Gondelach and Linares 2020[24]; Gershman 2026[25] | $\sigma$ 0.142, 8.4 % [−3.9, 18.3] lower (25 % required: not supported); 24.7 % lower in storms; as a forecast 16.8 % worse than JB2008; 3-day ratio 1.02; 0.270 on HASDM's windows | E8 |

**An open, DCA-form correction from public operator states** E13. HASDM's Dynamic Calibration

of the Atmosphere (DCA) corrects exospheric temperature every 3 hours from the drag of about 75–100 tracked calibration satellites, each with a constrained ballistic-coefficient correction[21]. E13 solves that form from public data alone: the common mode s of the daily drag of 36 Planet Doves, read through orbital energy, corrects JB2008's temperature, with a ballistic coefficient held constant per satellite over a block and a forecast with a 48-hour time constant. Published work derives density from operator ephemerides and scores density alone[26,27,28], or assimilates commercial GPS states[29]. E13 scores the correction on satellites that did not calibrate it, over ten months of Planet states against Swarm and GRACE-FO, with the storm of January 2026 in a held-out block. With x′ the logarithmic residual of observed against JB2008 density in a 12-hour window,

$$R_1 = 1 - \frac{\mathrm{sd}(x' - s)}{\mathrm{sd}(x')}$$

| Pre-registered test (threshold) | Test A, Jan–Mar 2026, Kp 8.7 storm | Test B, Apr–Jul 2026 |
|---|---|---|
| H1: Swarm and GRACE-FO density spread explained, $R_1$ ($\geq 0.25$) | 0.247 [0.137, 0.347]: not supported | 0.401 [0.283, 0.501]: supported |
| H3: held-out Doves' drag explained ($\geq 0.2$) | 0.378 [0.223, 0.477]: supported | 0.254 [0.135, 0.359]: supported |
| H4: Swarm position error ratio at 1 and 3 days ($\leq 0.9$) | 1.02, 0.92: not supported | 1.01, 1.00: not supported |
| H5: held-out Doves, the same | 0.98, 0.97: not supported | 1.00, 1.01: not supported |
| H6: control, correction shifted 7 days ($R_1 < 0.05$) | −0.149: passes | −0.082: passes |

The correction does not follow storms. Exploratory, on one NRLMSISE-00 reference for every split: the fleet explains 0.47–0.48 of NRLMSISE-00's larger spread, leaving residuals of 0.075–0.095 against 0.069–0.086 on JB2008.

### *5.5 Maneuver detection*

The detector compares the trajectories of consecutive element sets at one instant, whatever propagator answers its queries: a crossing test and a robust likelihood-ratio test for steps in orbit-averaged energy and orbit plane, merged by default[10]. It builds on element-set time series[30] and the generalized likelihood ratio for jumps[31]. Thresholds came from the ISS in 2020–2024; every other window was read once. A planned ISS burn counts as truth only when NASA's next trajectory shows it executed; one 2021 plan was not.

| Held-out class, 2025–2026 | Truth | Found | False alarms |
|---|---|---|---|
| ISS | 25 burns | 22; the misses are 0.1 m/s settling burns minutes before reboosts; median time error 10 min, $\Delta$v ratio 1.00 | 0.57 per year |
| Metop-SG A1 | 24 EUMETSAT maneuvers | 14; 12 of 16 at 0.1 m/s or more | 0 |
| GPS, 4 satellites | 6 maneuvers in precise orbits | 6 | 3.8 per object-year |
| GEO (Intelsat) | 11 post-maneuver files | 4 | 24 events per object-year; the truth lists one maneuver per satellite |

## 6 Uncertainty

Covariance propagates through HPOP's state-transition matrix $\Phi$ with a declared white-acceleration process noise Q[10]:

$$P(t) = \Phi(t, t_0)\, P_0\, \Phi(t, t_0)^{\mathsf{T}} + Q(t)$$

$\Phi$ supplies neither $P_0$ nor Q; they come from a fit, a provider or an empirical model. Realism is scored against independent precise orbits by $d^2 = \mathbf{e}^{\mathsf{T}} P^{-1} \mathbf{e}$ for a position error e, against $\chi^2$ with 3 degrees of freedom: a covariance is realistic when the mean of $d^2/3$ lies in [0.8, 1.25] and its 95 % coverage in [0.93, 0.97], the rule of E14 after Zaidi and Hejduk[32].

| Covariance | Prior art | Held-out result | Evidence |
|---|---|---|---|
| Formal covariance of an HPOP fit to laser ranges | Weighted least squares[16] | Too small: mean $d^2/3$ 16–21, coverage 0.14–0.18 | E6 |
| HPOP fit to OMM history alone | Levit and Marshall 2011[19] | Realistic for laser-ranged LEO (0.68–0.83, coverage 0.96–0.98); too small for the ISS within a day (4.8–7.6) | E6 |
| Batch fit to 3 days of OMM history (GPS) | Osweiler 2006[33]; Geul et al. 2017[34] | Mean $d^2/3$ 1.23 and 1.05 at 0 and 1 day, 81 at 3 days; consistency failed at every horizon | E2 |
| Errors of consecutive OMMs, against precise orbits | Autocorrelation shown by Osweiler[33] | Along-track correlation 0.61–1.00 in four regimes. Four published constructions[33,35,34,36] are overconfident: mean $d^2/3$ from 13 to over $10^7$. A covariance that models the correlation scores better at 1 day in three regimes but is consistent only for laser-ranged satellites at some horizons | E2b |
| SGP4 with the E2b error model, calibrated by regime and age | Zaidi and Hejduk 2016[32] | Realistic in 2 of 35 regime-horizon cells (11 on the selection window) | E14 |
| HPOP from the OMM epoch state, with $P_0$ and Q per regime | SP covariance as the Astrodynamics Support Workstation (ASW) forms it[37] | 5 of 35 (6 on the selection window): 500–800 km at epoch, above 800 km to 2 days | E14 |
| Density-forecast consider term, in the form of the Dynamic Consider Parameter | Hejduk 2019[38] | No measurable effect in storms; density-error coverage 0.83–0.86 against 0.95 | E14 |

On the held-out window, maneuvers, the storm of 4 July 2026 and GNSS repositioning put heavy tails on the errors: mean $d^2/3$ rises while medians and coverage stay near nominal in LEO, so realism found on the selection window did not carry over. Operator covariance enters as supplied, labeled with its source.

## 7 Dynamics and verification

HPOP's full force model, as the catalog and the conjunction screen run it[10], beside the SP options a Vector Covariance Message (VCM) names[39]:

| Model | HPOP | SP, as a VCM names it | Equivalence |
|---|---|---|---|
| Geopotential | EGM2008 to degree and order 70; EGM96 and tesseral truncation (36Z,24T) selectable | EGM-96 with zonal and tesseral degrees | EGM96 cases within 1 cm of Orekit |

| Model | HPOP | SP, as a VCM names it | Equivalence |
|---|---|---|---|
| Atmosphere | NRLMSISE-00 (Orekit's to 1e-8); JB2008 (Orekit's to 2e-14); Jacchia-Roberts (GMAT's[40], bit for bit) | JBH09, which is JB2008 with HASDM's calibration[41]; JAC70; MSIS90 | JB2008 matches Orekit's; HASDM's calibration is not public (section 5.4); Jacchia-Roberts and NRLMSISE-00 are declared substitutes for JAC70 and MSIS90 |
| Third bodies | Sun, Moon, Venus, Mars and Jupiter from DE440s | Sun and Moon | Sun and Moon within 1 cm of Orekit |
| Radiation | Cannonball with a conical shadow; Earth albedo and infrared | Solar radiation pressure, AGOM | Within 1 cm of Orekit |
| Tides, relativity | IERS 2010 solid tides and relativity; FES2004 ocean tides to 30×30 | Solid Earth tides | Within 1 cm of Orekit |
| Drag parameter, thrust | $Cd \cdot A/m$ and its rate; in-track acceleration | B, BDOT; in-track thrust | Within 1.6 cm of Orekit |
| Integrator | RK78 at 1e-12, steps of at most 300 s; its own error after 3 days 0.1–1.2 m | Eighth-order Gauss–Jackson on Cowell's equations[42,43] | Declared substitute, converged |
| Earth orientation | IAU 2006/2000A with daily IERS rows | IAU 1980 nutation, truncated | Declared substitute |

*Agreement with Orekit*

HPOP's Orekit 13.1 suite[44] holds 72 cases on five orbits (400 km, 700 km sun-synchronous, GPS, geostationary, Molniya), each 24 hours, compared hourly on identical states, constants, Earth orientation and space weather[10]:

| Force set | Largest 3D difference |
|---|---|
| Point mass | 0.6 mm |
| Gravity field, third bodies, radiation pressure, tides, relativity, in-track thrust | Under 1 cm |
| With drag (NRLMSISE-00 or JB2008, daily space weather, drag rate) | 1.6 cm |

State-transition matrices agree within 2e-5 and parameter sensitivities within 1.4e-4 (relative). What remains is integration error on both sides, about 1e-11 of the distance flown. Integrating Jacchia's diffusion equations numerically reproduces the closed forms of Jacchia-Roberts[45] to 1.6e-4.

*Accuracy against precise orbits*

Experiment V1 starts the propagator from ILRS orbits of four geodetic spheres (LAGEOS-1/2 and ETALON-1/2; 16 seeds) and IGS orbits of 32 GPS satellites (96 seeds), adds forces one set at a time, and compares each 72-hour run with the same product V1. GPS radiation pressure is a deliberately crude cannonball. Median (largest) 3D error:

| Force set | Spheres, 24 h | Spheres, 72 h | GPS, 24 h | GPS, 72 h |
|---|---|---|---|---|
| E-a: point mass | 46 km (396 km) | 110 km (1,213 km) | 25 km (40 km) | 75 km (170 km) |
| E-b: zonal harmonics to degree 20 | 1.3 km (2.2 km) | 3.4 km (8.7 km) | 2.1 km (6.5 km) | 6.0 km (94 km) |

| Force set | Spheres, 24 h | Spheres, 72 h | GPS, 24 h | GPS, 72 h |
|---|---|---|---|---|
| E-c: E-b with the Sun and Moon | 257 m (2.1 km) | 677 m (6.2 km) | 331 m (868 m) | 983 m (89 km) |
| E-d: E-c with radiation pressure | 246 m (2.1 km) | 670 m (6.2 km) | 277 m (661 m) | 853 m (89 km) |
| E-e: E-d with the full 20×20 field and IERS Earth orientation | 6.2 m (15 m) | 18.6 m (51 m) | 29.5 m (117 m) | 92 m (90 km) |
| E-f: E-e with IERS 2010 solid tides and relativity | 4.8 m (10 m) | 13.9 m (31 m) | 28.5 m (117 m) | 91 m (90 km) |

The GPS maxima beyond a day are one satellite, NORAD 35752, in every force set. Of four pre-registered checks, two passed and two failed:

1. Passed: the two-body return, 0.8 mm.
2. Passed: the E-d bound at 24 hours, 246 m against 500 m.
3. Failed: an added force did not reduce every seed's error (14 violations, all without tesseral harmonics).
4. Failed: a 20×20 field without the Sun and Moon beat zonals alone for only 53 of 96 GPS seeds.

HPOP's resident path, which holds the catalog in memory, reproduces its execution path to 0.052 mm over four days, at 0.645 s per object-day on one thread[10].

## 8 Association and selection

Association asks whether records describe one object; selection picks a solution for an epoch, interval and use; estimation derives a new one. Candidates come from provider crosswalks, international designators and native-identifier histories, Vimpel's datefirst fields included; names and bare numbers are not join keys. Compatibility is tested on normalized ephemerides over shared intervals. Each candidate link is compatible, rejected, ambiguous or insufficient, and accepted links stay reversible. Selection takes, per regime and horizon, the source with the lowest median error on a training window of at least 30 samples E3.

| Rule | Measured | Evidence |
|---|---|---|
| Selection by evidence | GPS: 3.5 cm at issue (section 5.1) | E3 |
| Fusion of sources | Never better than selection: 1.07–1.17 times its error in GPS at 1–7 days, equal elsewhere | E3 |
| Covariance intersection[46] of one provider's consecutive OMMs | Collapses onto one set (0.90–1.00 of selection's error); naive fusion is overconfident (mean $d^2/3$ 3.3–700) | E2b |

## 9 SP message interchange

The VCM carries an SP state, its force settings and an equinoctial covariance[39]. SDN's adapter reads a VCM into an HPOP request and writes the result back as a VCM, transforming the covariance with an exact Jacobian computed by forward-mode dual numbers[47]. The format states no covariance units, and no public definition was found. Four messages settle them:

1. The survey's ISS sample
2. An SP message for a geostationary orbit
3. An SP message for a GPS orbit
4. An SP message for an eccentric (e = 0.59) orbit

| Covariance rows | Reading | Evidence |
|---|---|---|
| Mean motion | Relative, dn/n; covariance scaled by max(1, weighted RMS)² | Every printed U, V and W sigma reproduced within 1 %; no absolute unit does (radians per 1000 s misses two radial sigmas by 1.7 and 2.0 times) |
| B and AGOM | Fractions of the coefficients (1.5–9.7 %) | The GPS message against ESA's final orbit for one day: radial and in-track RMS 0.46 and 0.64 of their sigmas, cross-track 1.27; read as printed, the sigmas overstate the error 7 to 20 times |

A round trip through VCM text in SP's number format recovers the covariance to 2e-5 of its sigmas. A VCM naming JBH09 runs as JB2008 on SET's indices, which the caller supplies.

## 10 Publication and reproduction

Publication is transactional: acquire and hash the bytes, verify, normalize, validate, sign a manifest, then announce the new catalog head. Consumers read one complete, immutable snapshot; each update names the object, the prior and new solutions, the epoch, the recipe and the provenance. Content addressing and independent replication keep a snapshot retrievable from any node that holds it.

An accepted numerical solution can carry an SGP4 companion: mean elements fitted to it over a declared interval and scored on a held-out stretch before release, as CelesTrak's supplemental GP fits operator ephemerides[48] and the 18th Space Defense Squadron's eGP fits SP states[49]. The fit RMS that CelesTrak publishes for each set is the benchmark.

Each experiment's numbers come from a run manifest: the repository and module commits, each module's WebAssembly SHA-256 and the SHA-256 of every input; reports are generated from metrics[1]. The experiments' public site runs the same modules in a web browser: a V1 seed reproduces the committed result exactly, and any Orekit case re-runs from its exact request. Space-Track element sets, SpaceX and Planet files and SET's indices are cited by URL and SHA-256, not copied.

| Evidence | Commit |
|---|---|
| Experiments V1, E1–E8, E13 and E14 | orbit-accuracy-experiments `357d992d6c62daac4ccbb2fd230634ac292192b8` |
| HPOP 1.3.0, its Orekit suite, screen and covariance calibration; maneuver detection 0.3.0 | space-data-network-modules `d4a0b1155f64ac088694f9758c32e3b691952674` |
| HPOP in V1 | space-data-network-modules `750b6399173a2678b8e8d0885557976d17a9ad5c` |
| VCM adapter | space-data-network-modules `1879d8c5c17c061e157c100723a8423c59fd98e2` |
| Vimpel conversion and refinement | space-data-network-modules `49e159d7003cac3d3e65b03170f11909fa86dd2c` |

HPOP's WebAssembly binary in V1, SHA-256:

`473d527175de6606ccf305c45b37218288634fa9de2dcab0d72564a190bdabd1`

## 11 Measured accuracy

Median 3D position error against precise orbits, by object class and age of the product:

| Class (objects) | Product | At issue | 1 day | 3 days | 7 days | Evidence |
|---|---|---|---|---|---|---|
| GPS (32) | Ultra-rapid orbit, propagated | 3.5 cm | 30 m | 104 m | 336 m | E3 |
| GPS (32) | Public OMM, SGP4 | 1.57 km | 1.61 km | 1.68 km | 1.53 km | E3 |
| Laser-ranged LEO (5) | HPOP fitted to laser ranges | 7 m | 9 m | 17 m | 42 m | E6 |
| Laser-ranged LEO (5) | Public OMM, SGP4 | 472 m | 485 m | 482 m | 685 m | E6 |
| Laser-ranged LEO, 800–1,500 km (5) | Public OMM, SGP4 | 314 m | 326 m | 326 m | 569 m | E3 |
| Laser-ranged MEO (5) | Public OMM, SGP4 | 411 m | 438 m | 395 m | 413 m | E3 |
| LEO, 400–800 km (6) | Public OMM, SGP4 | 566 m | 597 m | 760 m | 1.24 km | E3 |

The HAC's position errors are not published[2]. The public figures: element sets in the HAC's own LEO conjunction regime (perigee above 500 km) err by 0.6–3 km from epoch to 72 hours[49]; one ASW covariance shown as an illustration has $\sigma$ of 8.3 m radial, 332 m in-track and 2.1 m cross-track at epoch[37]; HASDM's density spread is 0.078 against JB2008's 0.148 (section 5.4).

## 12 What conjunction assessment consumes

Conjunction screening[50] takes from one catalog snapshot, per object:

| Input | From the catalog |
|---|---|
| Trajectory | The OMM for SGP4, or one propagator's trajectory; one propagator per run |
| Covariance | Supplied by the operator, fitted, or modeled (section 6), with its calibration label |
| Hard-body radius | Supplied, half the catalog size, from radar cross-section, or the default, with its basis |
| Provenance | Snapshot, source records and recipe |

An OMM supplies no covariance. A collision probability needs relative-state uncertainty with independent calibration and stated correlation, geometry and hard-body radius; without it the screen reports only the covariance-free maximum probability.

## Appendix A. Terms

| Term | Meaning |
|---|---|
| ASO | Anthropogenic space object |
| OMM, GP | CCSDS Orbit Mean-elements Message: SGP4 mean elements, also distributed in the legacy two-line format; general perturbations element set |
| SGP4, B* | The analytic propagator of OMMs; its drag term |
| HPOP | High-precision orbit propagation: numerical integration of the full force model of section 7 |
| Provider claim | An attributed product that already embeds a provider's models and inference |
| Level 0 to 3 | Processing levels of orbit data (section 2) |
| Ultra-rapid, final orbits | GNSS analysis-centre orbits issued in near real time with a prediction; definitive orbits issued about two weeks later |
| SLR, ILRS | Satellite laser ranging; the International Laser Ranging Service |
| TEME, EME2000, GCRF, ITRF | SGP4's output frame; the J2000 mean equator and equinox; the Geocentric Celestial Reference Frame; the terrestrial frame |

| Term | Meaning |
|---|---|
| $d^2$, mean $d^2/3$ | Squared Mahalanobis distance of an error under its covariance; the mean of $d^2/3$ is 1 for a realistic covariance |
| DCA, DCP | HASDM's Dynamic Calibration of the Atmosphere; the Dynamic Consider Parameter for density-forecast error |
| HASDM, JB2008, JBH09 | High Accuracy Satellite Drag Model; the Jacchia-Bowman 2008 density model; JB2008 with HASDM's calibration |
| SP, HAC, ASW | Special Perturbations orbit determination and propagation; the High Accuracy Catalog; the Astrodynamics Support Workstation |
| VCM | SP Vector Covariance Message |
| Supplemental GP, eGP | CelesTrak's element sets fitted to operator ephemerides; the 18th Space Defense Squadron's element sets fitted to SP states |
| AOE | SpaceMapper's catalog of element sets from its own sensors |
| Covariance intersection | A fusion rule that stays consistent under unknown correlation |

**Data credits**

- IGS products, courtesy of the International GNSS Service and its analysis and data centres (Johnston, Riddell and Hausler 2017, doi:10.1007/978-3-319-42928-1). © Navigation Support Office at ESA/ESOC 2026.
- GFZ rapid MGEX orbits of the IGS Multi-GNSS Experiment (Montenbruck et al. 2017, doi:10.1016/j.asr.2017.01.011), obtained from NASA CDDIS (Noll 2010, doi:10.1016/j.asr.2010.01.018).
- ILRS data and products (Pearlman et al. 2019, doi:10.1007/s00190-019-01241-1), from the EUROLAS Data Center, DGFI-TUM (Noll et al. 2019, doi:10.1007/s00190-018-1207-2). NERC Space Geodesy Facility orbits; contains British Geological Survey materials © UKRI 2026, Open Government Licence v3.0.
- Contains modified Copernicus Sentinel data 2026.
- © ESA 2026; data provided by the European Space Agency: SW_SP3xCOM_2_ (doi:10.57780/esa-8579355), SW_DNSxPOD_2_ (doi:10.57780/esa-ef59315), GF_DNSxACC_2_ (doi:10.57780/esa-2559fd1).
- Licata, Mehta, Tobiska, Bowman and Pilinski (2021), CHAMP and GRACE-A density estimates with associated HASDM and JB2008 predictions, Zenodo, doi:10.5281/zenodo.4602380, CC BY 4.0.
- Source: USSPACECOM / 18th Space Defense Squadron, via Space-Track.org.
- SatNOGS Network observations (Libre Space Foundation; the station and observer named in each record), https://network.satnogs.org/, CC BY-SA 4.0.
- SpaceX Starlink public ephemerides, https://api.starlink.com/public-files/ephemerides/ (SpaceX). Planet Labs PBC orbital ephemerides, https://ephemerides.planet-labs.com/, © Planet Labs PBC, CC BY-NC 4.0. ISS trajectory: NASA JSC Flight Operations Directorate (U.S. Government work). China Manned Space Engineering Office, https://www.cmse.gov.cn/. Intelsat (SES) public ephemerides, https://my.intelsat.com/ephemeris/public. EUMETSAT.
- JSC Vimpel, http://spacedata.vimpel.ru/. SpaceMapper, https://spacemapper.cn/. CelesTrak, https://celestrak.org (T. S. Kelso). Data from J. McDowell, planet4589.org.
- The JB2008 solar and geomagnetic indices are provided for scientific use courtesy of Space Environment Technologies (https://spacewx.com/jb2008/). Kp, ap, Ap: GFZ Helmholtz Centre for Geosciences, CC BY 4.0 (Matzka et al. 2021, doi:10.5880/Kp.0001). Sunspot number: WDC-SILSO, Royal Observatory of Belgium, Brussels, CC BY-NC 4.0. F10.7: Dominion Radio Astrophysical Observatory and Natural Resources Canada. NOAA Space Weather Prediction Center (U.S. Government material).
- IERS EOP 20 C04 (IERS Earth Orientation Centre, Observatoire de Paris); finals2000A (IERS Rapid Service/Prediction Centre, U.S. Naval Observatory). JPL planetary ephemeris DE440 (Park et al. 2021, doi:10.3847/1538-3881/abd414), via NASA/JPL NAIF. EGM2008: NGA (Pavlis et al. 2012, doi:10.1029/2011JB008916). EGM96: NASA GSFC and NIMA (Lemoine et al. 1998). FES2004 (Lyard et al. 2006) as distributed with the IERS Conventions (2010).

## References


*R1*

Edgesource. Orbit accuracy experiments: frozen plans, run manifests, generated reports and data licenses, with a browser site that re-runs them. Commit 357d992d6c62daac4ccbb2fd230634ac292192b8. Repository, site

*R2*

Edgesource. HAC data sources: 97 public and login-only sources, their access, terms and use, and what is public about the HAC. Inventory

*R3*

Licata, R. J., Mehta, P. M., Tobiska, W. K., Bowman, B. R., and Pilinski, M. D. CHAMP and GRACE-A density estimates with associated HASDM and JB2008 predictions. Zenodo, 2021. doi:10.5281/zenodo.4602380

*R4*

SpaceMapper. AOE catalog and orbital database. AOE catalog

*R5*

JSC Vimpel and Keldysh Institute of Applied Mathematics. Orbit parameters of newly detected HEO space debris objects. Bulletin

*R6*

Vallado, D. A. Fundamentals of Astrodynamics and Applications, 4th edition. Microcosm Press, 2013.

*R7*

Edgesource. Vimpel epoch conversion, refinement study and 64-object audit. SDN modules repository, commit 49e159d7003cac3d3e65b03170f11909fa86dd2c, `analysis/catalog-composer/docs/`. Private repository, available on request (tj@edgesource.com).

*R8*

Vallado, D. A., Crawford, P., Hujsak, R., and Kelso, T. S. Revisiting Spacetrack Report #3. AIAA 2006-6753, 2006. Paper

*R9*

Hoots, F. R., and Roehrich, R. L. Models for Propagation of NORAD Element Sets. Spacetrack Report No. 3, 1980. Report

*R10*

Edgesource. SDN modules repository (space-data-network-modules), commit d4a0b1155f64ac088694f9758c32e3b691952674: `propagator/hpop/README.md` (force model, Orekit suite, resident path, catalog cost), `analysis/gp-error-model/docs/hpop-calibration-2026-08.md`, `analysis/conjunction-assessment/docs/hpop-full-force-screen-2026-10.md`, `analysis/maneuver-detection/README.md`, `docs/studies/maneuver-detection.md`. Private repository, available on request (tj@edgesource.com).

*R11*

Petit, G., and Luzum, B., editors. IERS Conventions (2010). IERS Technical Note 36. Technical note

*R12*

Johnston, G., Riddell, A., and Hausler, G. The International GNSS Service. Springer Handbook of Global Navigation Satellite Systems, 967–982, 2017. Products

*R13*

Pearlman, M. R., et al. The ILRS: approaching 20 years and planning for the future. Journal of Geodesy 93, 2161–2180, 2019. doi:10.1007/s00190-019-01241-1

*R14*

Dyreby, K., Caldas, F., and Soares, C. Analyzing Data Quality and Decay in Mega-Constellations: A Physics-Informed Machine Learning Approach. arXiv:2510.11242, 2025. Preprint

*R15*

Jankovic, D. How long can you trust a Starlink TLE? An empirical comparison of SGP4 and high-fidelity propagation against operator-updated truth across a megaconstellation. arXiv:2605.19850, 2026. Preprint

*R16*

Tapley, B. D., Schutz, B. E., and Born, G. H. Statistical Orbit Determination. Elsevier Academic Press, 2004.

*R17*

Ly, D., Lucken, R., and Giolito, D. Correcting TLEs at epoch: application to the GPS constellation. Journal of Space Safety Engineering 7(3), 2020.

*R18*

Hallgarten La Casta, M. I., and Amato, D. Generalisable bias correction of two-line element sets in the medium Earth orbit regime. 9th European Conference on Space Debris, 2025.

*R19*

Levit, C., and Marshall, W. Improved orbit predictions using two-line elements. Advances in Space Research 47(7), 1107–1115, 2011. arXiv:1002.2277

*R20*

Bowman, B. R., Tobiska, W. K., Marcos, F. A., Huang, C. Y., Lin, C. S., and Burke, W. J. A New Empirical Thermospheric Density Model JB2008 Using New Solar and Geomagnetic Indices. AIAA 2008-6438, 2008. doi:10.2514/6.2008-6438

*R21*

Storz, M. F., Bowman, B. R., Branson, J. I., Casali, S. J., and Tobiska, W. K. High accuracy satellite drag model (HASDM). Advances in Space Research 36(12), 2497–2505, 2005. doi:10.1016/j.asr.2004.02.020

*R22*

Doornbos, E. Empirical Model Calibration. In Thermospheric Density and Wind Determination from Satellite Dynamics, Springer Theses, 2012. doi:10.1007/978-3-642-25129-0_6

*R23*

Picone, J. M., Emmert, J. T., and Lean, J. L. Thermospheric densities derived from spacecraft orbits: accurate processing of two-line element sets. Journal of Geophysical Research 110(A3), 2005. doi:10.1029/2004JA010585

*R24*

Gondelach, D. J., and Linares, R. Real-time thermospheric density estimation via two-line element data assimilation. Space Weather 18(2), e2019SW002356, 2020. doi:10.1029/2019SW002356

*R25*

Gershman. Near Real-Time Thermospheric Density Monitoring from Public Satellite Tracking and Space Weather Data. NASA, 2026. doi:10.64631/vfnq1575

*R26*

Ou, Zhong, Hao, Li, et al. Near-Real-Time Global Thermospheric Density Variations Unveiled by Starlink Ephemeris. Remote Sensing 17(9), 1549, 2025. doi:10.3390/rs17091549

*R27*

Yamamoto. Tomography of thermospheric density from Starlink Ephemeris: initial report. Earth, Planets and Space 78, 2026. doi:10.1186/s40623-026-02509-5

*R28*

Fitzpatrick, Sutton, Pilinski, and Palo. Harnessing Satellite Constellations as Signals of Opportunity for an Enhanced Specification of the Satellite Drag Environment in the Critical LEO Regime. Space Weather, 2026. doi:10.1029/2025SW004611

*R29*

Gondelach, D. J., and Linares, R. Real-Time Thermospheric Density Estimation via Radar and GPS Tracking Data Assimilation. Space Weather 19, e2020SW002620, 2021. doi:10.1029/2020SW002620

*R30*

Lemmens, S., and Krag, H. Two-Line-Elements-Based Maneuver Detection Methods for Satellites in Low Earth Orbit. Journal of Guidance, Control, and Dynamics 37(3), 860–868, 2014. doi:10.2514/1.61300

*R31*

Willsky, A. S., and Jones, H. L. A generalized likelihood ratio approach to the detection and estimation of jumps in linear systems. IEEE Transactions on Automatic Control 21(1), 108–112, 1976. doi:10.1109/TAC.1976.1101146

*R32*

Zaidi, W. H., and Hejduk, M. D. Earth Observing System Covariance Realism. AIAA/AAS Astrodynamics Specialist Conference, AIAA 2016-5628, 2016. doi:10.2514/6.2016-5628

*R33*

Osweiler, V. P. Covariance estimation and autocorrelation of NORAD two-line element sets. MS thesis, Air Force Institute of Technology, AFIT/GSS/ENY/06-M09, 2006. Thesis

*R34*

Geul, J., Mooij, E., and Noomen, R. TLE uncertainty estimation using robust weighted differencing. Advances in Space Research 59(10), 2522–2535, 2017. doi:10.1016/j.asr.2017.02.038

*R35*

Thompson, Gossner, Sais, and Cunningham. Fast covariance propagation for two-line element sets. AMOS Technical Conference, 2019. doi:10.64861/twqd5985

*R36*

Verhaeghe, De Clerk, Lauwens, Delabie, Vandepitte, and Vandenbussche. Predicting TLE covariance using transformer-based time series forecasting. Small Satellite Conference, SSC25-IV-03, 2025. Paper

*R37*

Hejduk, M. D. ASW Covariance Introduction and Formation. NASA, 2016. NTRS 20160005042

*R38*

Hejduk, M. D. Three CA-Related Covariance Issues and Their Solutions. NASA, 2019. NTRS 20190026790

*R39*

Space Data Standards. Legacy message survey: the Vector Covariance Message format and sample; PRW and VCM schemas. Survey

*R40*

NASA Goddard Space Flight Center. General Mission Analysis Tool (GMAT), atmosphere models. Project

*R41*

Pachura, D., and Hejduk, M. D. Atmospheric Model at JSpOC. NASA, 2016. NTRS 20160005044

*R42*

National Research Council. Continuing Kepler's Quest: Assessing Air Force Space Command's Astrodynamics Standards. The National Academies Press, 2012. doi:10.17226/13456

*R43*

Berry, M. M., and Healy, L. M. Implementation of Gauss–Jackson Integration for Orbit Propagation. Journal of the Astronautical Sciences 52(3), 331–357, 2004. Paper

*R44*

CS GROUP. Orekit 13.1, open-source space flight dynamics library, Apache License 2.0. Project

*R45*

Roberts, C. E., Jr. An analytic model for upper atmosphere densities based upon Jacchia's 1970 models. Celestial Mechanics 4, 368–377, 1971. doi:10.1007/BF01231398

*R46*

Julier, S. J., and Uhlmann, J. K. A non-divergent estimation algorithm in the presence of unknown correlations. Proceedings of the American Control Conference, 2369–2373, 1997.

*R47*

Edgesource. VCM adapter (`analysis/vcm-adapter`). SDN modules repository, commit 1879d8c5c17c061e157c100723a8423c59fd98e2. Private repository, available on request (tj@edgesource.com). Parity record: vcm-parity.md

*R48*

CelesTrak. Supplemental GP element sets: methodology for fitting operator ephemerides with SGP4. Supplemental GP

*R49*

NASA. Spacecraft Conjunction Assessment and Collision Avoidance Best Practices Handbook, Vol. 2, Technical Appendices, August 2026. NTRS 20260000457

*R50*

Koury, A. Fast All-vs-All Conjunction Screening. Space Data Network technical whitepaper. Paper